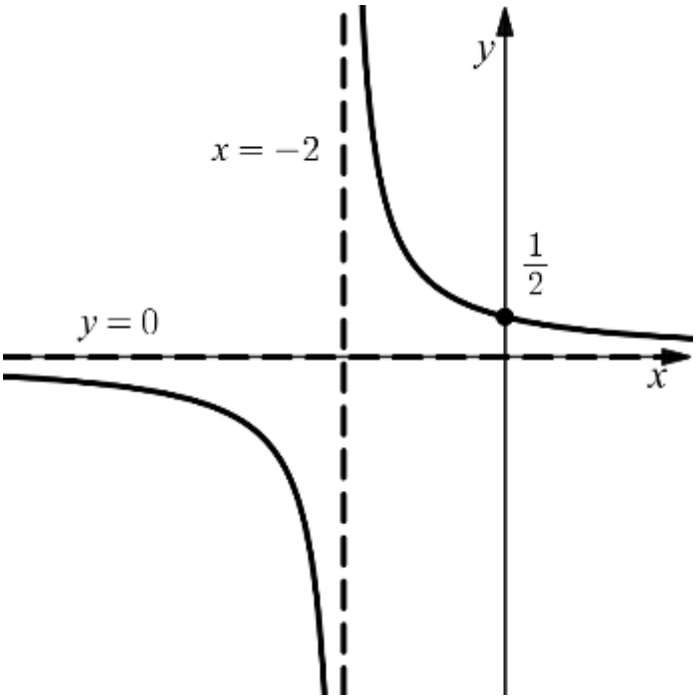
2.59 Reciprocal Functions: Inequalities

Worked Solutions

1 a)

Asymptotes: $x = -2, y = 0$

When $x = 0, y = \frac{1}{2}$



$\{x : x \in \mathbb{R}, x < -2\}$

2 a)

$x \in [a, -\frac{1}{2}) \cup (1, b]$

2 b)

$x \in (-1, a] \cup [b, \frac{5}{2})$

2 c)

$x \in [a, -1) \cup (\frac{2}{3}, 3)$

3 a)

$$\frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)}$$

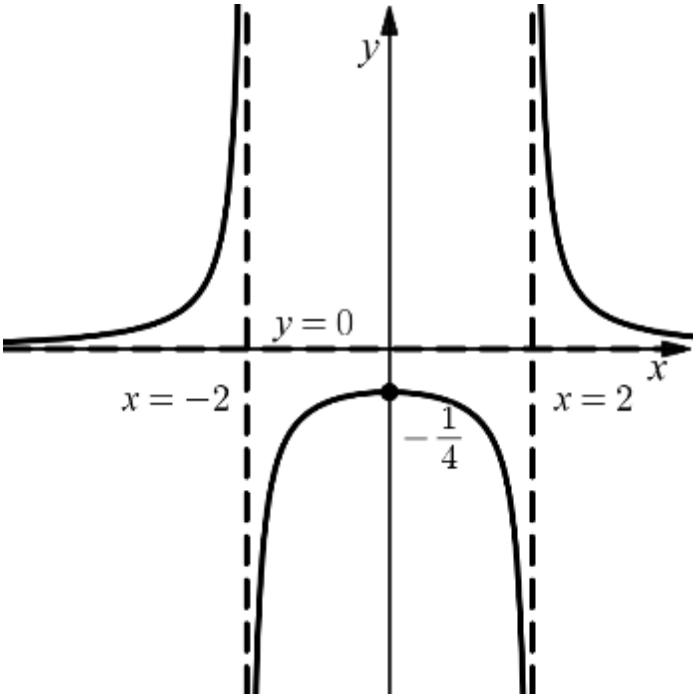
Asymptotes: $x = -2, x = 2$ and $y = 0$

Min at $(0, -4)$ becomes local max at $(0, -\frac{1}{4})$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -2$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -2$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$x \in (-2, 2)$

3 b)

$$\frac{1}{6+5x-x^2} = \frac{1}{-(x+1)(x-6)}$$

When $x = 0, y = \frac{1}{6}$

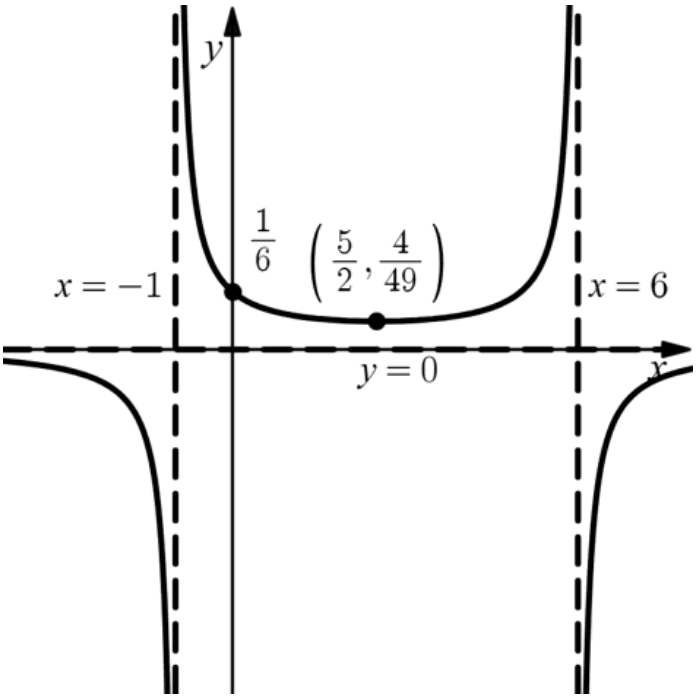
Asymptotes: $x = -1, x = 6$ and $y = 0$

Max at $(\frac{5}{2}, \frac{49}{4})$ becomes local max at $(\frac{5}{2}, \frac{4}{49})$

As $x \to -\infty, \frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty, \frac{1}{\mathrm{f}(x)} \to 0^-$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 6$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 6$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$



$x \in (-\infty, -1) \cup (6, \infty)$

4 a)

$$\frac{1}{x^2-9} = \frac{1}{(x-3)(x+3)}$$

When $x = 0, y = -\frac{1}{9}$

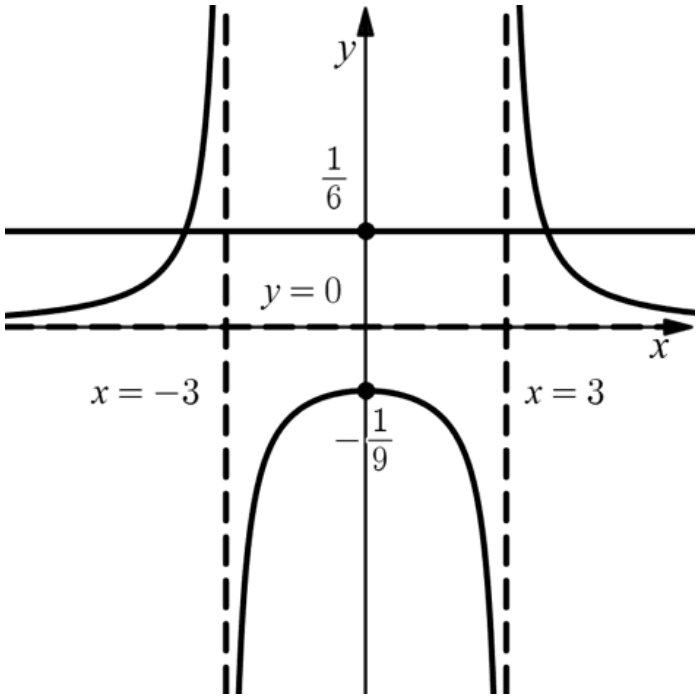
Asymptotes: $x = -3, x = 3$ and $y = 0$

Min at $(0, -9)$ becomes local max at $(0, -\frac{1}{9})$

As $x \to -\infty, \frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty, \frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -3$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -3$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 3$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 3$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$$\frac{1}{6} = \frac{1}{x^2-9} \Rightarrow x^2 - 9 = 6 \Rightarrow x^2 - 15 = 0 \therefore x = \pm\sqrt{15}$$

$$\{x : x \in \mathbb{R}, -\sqrt{15} < x < -3\} \cup \{x : x \in \mathbb{R}, 3 < x < \sqrt{15}\}$$

4 b)

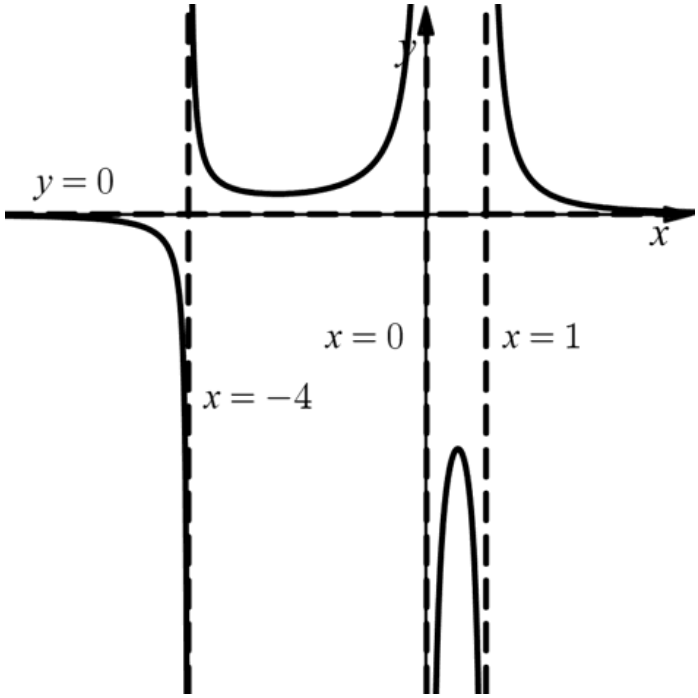
Asymptotes: $x = -4, x = 0$ and $x = 1$

As $x \to -\infty, \frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty, \frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -4$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -4$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$$\{x : x \in \mathbb{R}, x < -4\} \cup \{x : x \in \mathbb{R}, 0 < x < 1\}$$

5 a)

$$\mathrm{f}\left(\frac{1}{4}\right) = 12\left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^2 - 9\left(\frac{1}{4}\right) + 2 = \frac{3}{16} + \frac{1}{16} - \frac{9}{4} + 2 = 0$$

$\therefore (4x - 1)$ is a factor of $\mathrm{f}(x) = 12x^3 + x^2 - 9x + 2$

5 b)

$$12x^3 + x^2 - 9x + 2 = (4x-1)(3x^2 + x - 2) = (4x-1)(3x-2)(x+1)$$

$$\frac{1}{12x^3 + x^2 - 9x + 2} = \frac{1}{(4x-1)(3x-2)(x+1)}$$

When $x = 0, y = \frac{1}{2}$

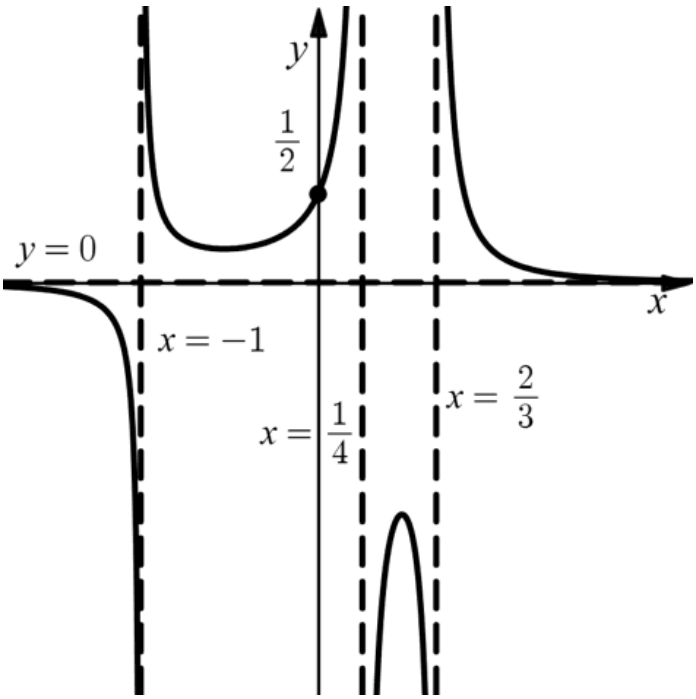
Asymptotes: $x = -1, x = \frac{1}{4}, x = \frac{2}{3}$ and $y = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to \frac{1}{4}$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to \frac{1}{4}$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to \frac{2}{3}$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to \frac{2}{3}$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



5 c)

$$\{x : x \in \mathbb{R}, x < -1\} \cup \{x : x \in \mathbb{R}, \frac{1}{4} < x < \frac{2}{3}\}$$

6 a)

$$\mathrm{f}\left(\frac{1}{3}\right) = 6\left(\frac{1}{3}\right)^3 - 5\left(\frac{1}{3}\right)^2 - 2\left(\frac{1}{3}\right) + 1 = \frac{2}{9} - \frac{5}{9} - \frac{2}{3} + 1 = 0$$

$\therefore (3x-1)$ is a factor of $\mathrm{f}(x) = 6x^3 - 5x^2 - 2x + 1$

6 b)

$$6x^3 - 5x^2 - 2x + 1 = (3x-1)(2x^2 - x - 1) = (3x-1)(2x+1)(x-1)$$

$$\frac{1}{6x^3 - 5x^2 - 2x + 1} = \frac{1}{(3x-1)(2x+1)(x-1)}$$

When $x = 0, y = 1$

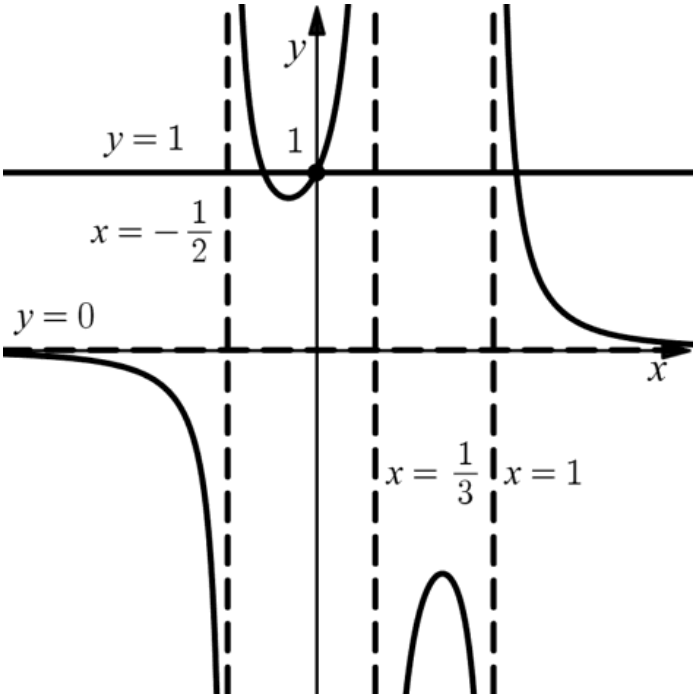
Asymptotes: $x = -\frac{1}{2}, x = \frac{1}{3}, x = 1$ and $y = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -\frac{1}{2}$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -\frac{1}{2}$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to \frac{1}{3}$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to \frac{1}{3}$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



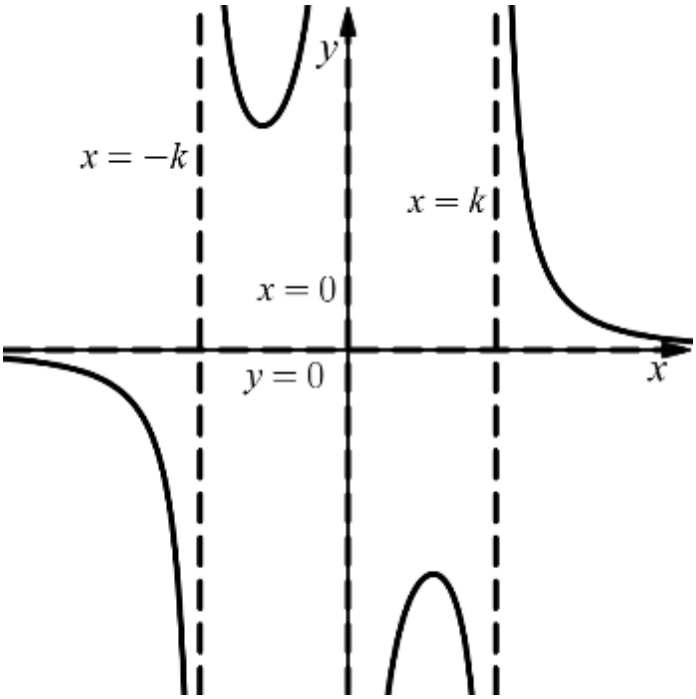
6 c)

$$\frac{1}{6x^3 - 5x^2 - 2x + 1} = 1 \Rightarrow 1 = 6x^3 - 5x^2 - 2x + 1 \Rightarrow 6x^3 - 5x^2 - 2x = 0$$

$$\Rightarrow x(6x^2 - 5x - 2) = 0 \therefore x = 0, x = \frac{5}{12} \pm \frac{\sqrt{73}}{12}$$

$$x \in (-\infty, -\frac{1}{2}) \cup (\frac{5}{12} - \frac{\sqrt{73}}{12}, 0) \cup (\frac{1}{3}, 1) \cup (\frac{5}{12} + \frac{\sqrt{73}}{12}, \infty)$$

7 a)



7 b)

$$\{x : x \in \mathbb{R}, -k < x < 0\} \cup \{x : x \in \mathbb{R}, x > k\}$$